python™ Libraries

FOR **BUILDING** Generative **AI Apps**





By SHAILESH SHAKYA @BEGINNERSBLOG.ORG

1. Transformers

Your AI toolbox, pre-packed with powerful language models.

- Simplifies access to state-of-the-art models like GPT, BERT, T5, etc.
- Provides pipelines for common NLP tasks: text generation, translation, summarization.
- Abstracts away complex model architectures, making them user-friendly.
- Enables easy fine-tuning on custom datasets for specific needs

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1. Transformers: example ↘

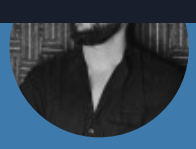
```python
from transformers import pipeline

# Text generation using GPT-2
generator = pipeline('text-generation', model='gpt2')
print(generator("Once upon a time", max_length=50))  # Generates a continuation of the given prompt

# Translation from English to French using T5
translator = pipeline('translation_en_to_fr', model='t5-small')
print(translator("Hello, world!"))  # Translates to "Bonjour, le monde!"

# Summarization using BART
summarizer = pipeline('summarization', model='facebook/bart-large-cnn')
print(summarizer("This is a long article about AI..."))  # Produces a concise summary
```

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2. PEFT

Efficiency booster for your AI models

- Fine-tunes large language models with less computational resources
- Leverages techniques like LoRA (Low-Rank Adaptation) to reduce memory footprint
- Makes it feasible to adapt powerful models on commodity hardware
- Accelerates training and inference times

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2. PEFT: Example ⤵

```python
from peft import PeftModel, LoraConfig, get_peft_model
from transformers import AutoModelForCausalLM, AutoTokenizer

# Load a pre-trained language model and its tokenizer
model = AutoModelForCausalLM.from_pretrained("bigscience/bloom-560m")
tokenizer = AutoTokenizer.from_pretrained("bigscience/bloom-560m")

# Configure PEFT (Lora) for efficient fine-tuning
peft_config = LoraConfig(task_type="CAUSAL_LM", inference_mode=False, r=8, lora_alpha=32, lora_dropout=0.1)
model = get_peft_model(model, peft_config)

# Fine-tune the model on your data (not shown here for brevity)
```

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3. Diffusers:

Your gateway to generating images and other media

- Implements various diffusion models for high-quality image generation
- Supports text-to-image, image-to-image, and inpainting tasks
- Provides pre-trained models and easy customization options
- Makes creating visually appealing content accessible

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3. Diffusers: Example ⤵

```python
from diffusers import
    StableDiffusionPipeline

# Load a pre-trained Stable Diffusion
    pipeline
pipe = StableDiffusionPipeline
    .from_pretrained("runwayml/stable
    -diffusion-v1-5")

# Generate an image from a text prompt
prompt = "a photorealistic image of a
    cat wearing a hat"
image = pipe(prompt).images[0]

# Save the generated image
image.save("cat_with_hat.png")
```

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4. LangChain

The architect for building conversational AI

- Chains together multiple language models and tools for complex tasks
- Manages memory and context for multi-turn conversations
- Integrates with external data sources for knowledge-grounded responses
- Provides a framework for building chatbots, question-answering systems, etc

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4. LangChain: Example ↘

```
1  from langchain.chains import
       ConversationChain
2  from langchain.llms import OpenAI
3
4  # Initialize a conversation chain using
       OpenAI's GPT-3
5  llm = OpenAI(temperature=0.9)  # Adjust
       temperature for creativity
6  conversation = ConversationChain(llm
       =llm)
7
8  # Interact with the chatbot
9  print(conversation.predict(input="Hi
       there!"))
10 print(conversation.predict(input
       ="What's your name?"))
```

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5. LlamaIndex

Your search engine for large language models

- Indexes and structures your data for efficient retrieval
- Enables querying large language models even if they can't fit in memory
- Supports various data sources and indexing techniques
- Helps build applications that leverage knowledge from vast datasets

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5. LlamaIndex: Example ↘

```python
from llama_index import
    SimpleDirectoryReader,
    GPTVectorStoreIndex, LLMPredictor
from langchain.llms import OpenAI

# Load data from a directory
documents = SimpleDirectoryReader
    ('data').load_data()

# Create a vector store index using
    OpenAI's embeddings
index = GPTVectorStoreIndex(documents)

# Query the index using a language
    model
query_engine = index.as_query_engine()
response = query_engine.query("What are
    the key takeaways from these
    documents?")
print(response)
```

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6. Chat UI & Gradio ⤵

(See the LangChain example above for a basic chatbot implementation. Enhance it with Chat UI for a visually appealing and interactive interface)

- Gradio: Democratize your AI models with user-friendly web interfaces
- Creates interactive demos for your machine learning models
- Supports various input and output types (text, images, audio, etc.)
- Requires minimal coding to get started
- Makes it easy to share your models with others

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6. Chat UI & Gradio: Example ⤵

```
1  import gradio as gr
2
3  def greet(name):
4      return "Hello " + name + "!"
5
6  # Create a simple Gradio interface with
       a text input and output
7  iface = gr.Interface(fn=greet, inputs
       ="text", outputs="text")
8  iface.launch()  # Launch the interface
       in your web browser
```

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7. OpenLLM & OpenAI Python ⤵

OpenLLM: Streamline deployment and management of large language models

- Supports various open-source LLMs and frameworks
- Provides tools for quantization, optimization, and serving
- Simplifies the process of deploying models to production environments

- OpenAI Python: Your bridge to OpenAI's powerful API
  - Access a wide range of AI models and capabilities
  - Generate text, images, code, and more
  - Integrates seamlessly with other Python libraries

(Refer to the LangChain and PEFT examples for how to utilize OpenAI's models and fine-tune them for specific tasks. OpenLLM helps deploy and manage such models effectively)





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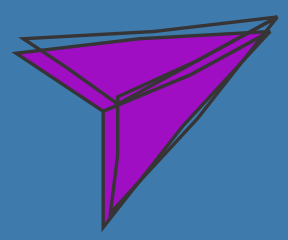
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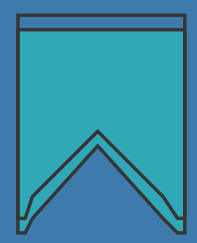
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